

MD. RESHAD AL MUTTAKI

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SUMMARY

EEE graduate from BRAC University with a research focus on semiconductor device design and simulation. Experienced in TCAD-based device modeling, VLSI design, digital-twin-driven optimization, and undergraduate teaching and mentorship, with six peer-reviewed Q1 journal publications and multiple manuscripts under review and expected to be published very soon. Former Student Tutor and Student Mentor in the Department of EEE, with a strong foundation in VLSI and mixed-signal design, committed to building a career in academia through excellence in teaching, impactful research, technological innovation, and inspiring and mentoring the next generation of engineers.

EDUCATION

Bachelor of Science in Electrical and Electronic Engineering BRAC University Relevant coursework in Electrical and Electronic Engineering	<i>Jan 2022 – Jan 2026</i> Major: Electronics CGPA: 3.81
Higher Secondary Certificate (HSC) Square School & College Rajshahi Board	<i>Passing Year: 2020</i> Major: Science GPA: 5.00
Secondary School Certificate (SSC) Square School & College Rajshahi Board	<i>Passing Year: 2018</i> Major: Science GPA: 5.00

PROFESSIONAL EXPERIENCE

Research Fellow <i>BRAC University</i>	<i>May 2026 – Present</i>
- Conducting research on “GaN-Based G4FET Technology: Fabrication, Characterization, Modeling, and Circuit Demonstration,” funded under the CREST initiative. - Designing a 5 nm AlGaN/GaN tri-gate FinFET (AS³-FinFET) in Silvaco Atlas TCAD, applying VLSI short-channel-effect theory and polarization-charge device physics. - Applying gate, doping, and spacer engineering principles to manage electrostatic scaling and interface effects at the sub-5 nm VLSI node. - Benchmarking the design against global FinFET/nanosheet device approaches, with a roadmap toward compact modeling and circuit-level demonstration.	
Research Assistant (RA) <i>BRAC University</i>	<i>Mar 2026 – Present</i>
- Conducting research on “Advancing Sustainable Biosensing: A Self-powered Biosensor for Real-time and Wireless Health Monitoring,” funded by RSGI of BRAC University. - Developing an e-skin-type biosensor in both simulation and hardware environments.	
Student Tutor (ST), BSRM SoE, Dept. of EEE <i>BRAC University</i>	<i>Jan 2024 – Oct 2025</i>
- Managed and mentored undergraduate students through tutorial and consultation sessions in both online and offline environments across multiple courses. - Prepared supplementary problem sets, worked examples, and exam-review materials to reinforce lecture content and address common conceptual gaps. - Held regular office hours and pre-exam review sessions, helping students build stronger fundamentals in circuit analysis, semiconductor devices, and energy-conversion topics.	
Student Mentor, FYAT, OAA <i>BRAC University</i>	<i>Jan 2024 – Jan 2025</i>
- Provided one-on-one mentorship to students as part of the First Year Advising Team (FYAT). - Guided first-year students on course selection, study habits, and adjustment to university life.	

COURSES TAUGHT AS STUDENT TUTOR (ST)

Electrical Circuits I (EEE/ECE101)

Summer 2024

- Foundational DC circuit analysis: Kirchhoff's laws, nodal/mesh analysis, Thevenin, Norton and superposition theorems, and transient response of RC/RL circuits.

Electronic Circuits I (EEE/ECE205)

Spring 2024

- Semiconductor device fundamentals and single-stage amplifier design covering diode, BJT, JFET, and MOSFET construction, characteristics, and circuit analysis.

Energy Conversion (EEE221)

Fall 2024 – Summer 2025

- Principles, construction, and performance analysis of DC and AC energy-conversion machines, including transformers, induction motors, and synchronous machines.

RESEARCH INTERESTS

1. Electronics & Semiconductor

- VLSI Design
- Semiconductor Devices
- Embedded Systems
- CAD Algorithms

2. AI & Cybernetics

- Artificial Intelligence & ML
- IoT & Cyber-Physical Systems
- Digital Twin Systems
- Computer Architecture

3. Biomedical Engineering

- Healthcare Technology
- Biomedical AI
- Flexible PCB & E-Skin
- Health & Bioinformatics

ACHIEVEMENTS AND HONORS

- **Vice Chancellor's (VC's) List** — Spring 2023; Summer 2023; Summer 2024
- **Dean's List** — Fall 2022; Fall 2023; Spring 2024; Fall 2024; Spring 2025

PUBLICATIONS

1. HERA: A Hierarchical Expected-Hypervolume Reinforcement-Guided Acquisition Framework and Digital Twin for Multi-Objective Optimization of Sub-3 nm Gate-All-Around FinFETs

Authors: Md. Reshad Al Muttaki, Tahmid Faiz Noor, SK Tahmed Salim Rafid

Status: Under Review in *Materials Today Electronics* | **1st Author**

A constrained multi-objective VLSI device-optimization framework for sub-3 nm gate-all-around FinFETs, developed to navigate the electrostatic and transport trade-offs inherent to aggressive gate-length scaling. The framework combines a Gaussian-process ensemble surrogate, a constrained expected-hypervolume acquisition function, and a reinforcement-learning-warm-started search to efficiently explore the device design space over a calibrated TCAD digital twin, with representative device designs cross-verified using DEVSIM drift-diffusion TCAD.

2. Towards smarter grids: A systematic review of wide area monitoring enhancing stability, protection, and promoting environmental sustainability

Authors: Md. Reshad Al Muttaki, Alvi Ibn Amzad Anil, Sadia Afrin, Shameem Hasan

Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **1st Author**

DOI: 10.1016/j.ecmx.2025.101424

3. AI-powered cybersecurity for smart grid communication: A systematic review of intrusion detection and threat mitigation systems

Authors: Sadia Afrin, Md. Reshad Al Muttaki, Alvi Ibn Amzad Anil, Shameem Hasan

Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **2nd Author**

DOI: 10.1016/j.ecmx.2025.101416

4. State-of-charge-aware charging opportunity detection for electric vehicles using data-driven learning and digital twin simulation

Authors: Md. Reshad Al Muttaki, Sadia Afrin, Alvi Ibn Amzad Anil, Shameem Hasan

Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **1st Author**

DOI: 10.1016/j.ecmx.2026.101775

5. **The next generation of energy storage for smart and sustainable power grids: A systematic review**

Authors: Alvi Ibn Amzad Anil, **Md. Reshad Al Muttaki**, Sadia Afrin, Shameem Hasan
Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **2nd Author**
DOI: 10.1016/j.ecmx.2026.101771

6. **Demand-based anomaly detection algorithm in electric vehicle charging systems using machine learning and digital twin simulation**

Authors: **Md. Reshad Al Muttaki**, Sadia Afrin, Alvi Ibn Amzad Anil, Shameem Hasan
Journal of Energy Storage | Q1, Top 6% | IF: 9.8 | CiteScore: 13.3 | **1st Author**
DOI: 10.1016/j.est.2026.122795

7. **Balancing cybersecurity, wireless charging, and safety: Digital twin-driven techno-economic analysis of electric vehicles**

Authors: Shameem Hasan, Abu M. Fuad, B.M. Khalid Hasan Drobo, **Md. Reshad Al Muttaki**, Sadia Afrin, Alvi Ibn Amzad Anil, Omar Farrok
Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **4th Author**
DOI: 10.1016/j.ecmx.2026.102107

CERTIFICATIONS

Industrial Training in VLSI Design

Oct 2025

Ulkasemi Pvt. Ltd.

- Analog Design — circuit-level design and analysis of analog building blocks.
- Digital Design — RTL-level design methodology for digital systems.
- Digital Verification — functional verification flows for digital circuit designs.
- Custom Layout — physical layout design for custom VLSI cells.
- Digital Circuit Design — design and implementation of digital circuit blocks.

SKILLS

Programming:	Python, C, MATLAB, Verilog, VHDL
EDA & Simulation:	Quartus, DEVSIM, Cadence, COMSOL Multiphysics, Silvaco TCAD, Altium Designer
Web & Markup:	HTML, CSS, LaTeX
Tools & Platforms:	Git & GitHub, Google Suite, MS Office Suite

REFERENCES

Dr. Mirza Rasheduzzaman

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SK Tahmed Salim Rafid

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